

Joosung (Benjamin) Park

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EDUCATION

Penn State University — BS in Computer Science; Minors in Mathematics & Art Expected May 2027

Goal: learn each culture, find the value to contribute — prioritize learning and applying machine learning

EXPERIENCE

Penn State College of Engineering, University Park, PA

Undergraduate Research Ambassador (CEOI) June 2025 – Present

- Connect students with faculty, programs, and research resources across Penn State campuses
- Advise peers on resume writing and interview preparation

Front Desk Student Worker - Engineering Student Support August 2025 – Present

- Manage in-person and hybrid student–advisor appointments, ensuring efficient check-ins

Academic Integrity Student Committee July 2026 – Present

- Serve as a student voice on academic integrity cases and policy within the College of Engineering

MC REU 2026 — Measuring Whether an AI Can Understand a Minecraft Builder's Intentions June – August 2026
from Raw Observation

- Built an end-to-end data pipeline: instrumented game capture, automated post-session processing, and human-in-the-loop labeling
- Trained a transformer decoder for next-token action prediction, reaching 71% top-1 accuracy and 59% eight tokens ahead
- Repurposed frozen foundation models (VPT, MineCLIP, Uni3D) from action generation to perception, training only lightweight heads to infer human intent

MC REU 2025 — Fine-Tuning LLMs for Code Smell Detection in C++ Game Development June – August 2025

- Fine-tuned two DeepSeek-Coder 1.3B models to detect code smells and generate automated fixes for open-source C++ game engine code
- Reduced GPU memory requirements with LoRA/QLoRA parameter-efficient fine-tuning, accelerating training iteration
- Developed a Python pipeline collecting and preprocessing GitHub C++ repositories into detection and fix-generation datasets

Honor Guard, Staff Sergeant, Republic of Korea Air Force August 2022 – June 2024

- Performed intricate drill routines with 100% precision at 100+ public events, national ceremonies, and state functions

PROJECTS

PathFinder — Yale Hackathon 2026 March 2026

- Engineered the AI/ML and backend of a satellite-image disaster-response tool in a two-person team: YOLO damage detection, FastAPI routing engine, geo-referencing pipeline

RippleJourney — ripple-journey.vercel.app March 2026

- Built a full-stack app in 3 days (Next.js, TypeScript, Prisma, PostgreSQL/pgvector) turning a raw research idea into a step-by-step roadmap with AI-matched faculty mentors via the Google Scholar API

LEADERSHIP & ACTIVITIES

Team Member, Builder's Collective — Penn State University: build with student founders and engineers 2026 – Present
at collegiate hackathons, shipping PathFinder at Yale Hackathon 2026

Mentor, Commonwealth Engineering Scholars Program — Penn State University: support incoming students through a smooth transition to University Park and encourage them to pursue research 2026 – Present

Member, Society of Hispanic Professional Engineers (SHPE) — Penn State University 2026 – Present

Member, Rowing Team (Crew) — Penn State University: train at daily 5:30 AM practices, learning the discipline and synchronized trust of crew 2025 – Present

Representative, International Student Organization: mentored 10+ new international students, bridging Korean and American campus cultures 2021 – 2022

Team Member, League of Legends Esports: analyzed opponents and set strategy with a five-player roster, winning the ECAC tournament 2021 – 2022